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Original Article

The social science side of marine biodiversity monitoring

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Abstract

Marine biodiversity monitoring is a complex task given the vast ocean areas to be covered, the types of data to be integrated, and the large number of possible parameters to consider when measuring biodiversity change. International agreements, such as the Convention on Biological Diversity (CBD) or the recently adopted 'BBNJ treaty'protecting biodiversity beyond national jurisdiction (BBNJ), encourage states to monitor marine life both within and beyond their national waters to foster marine biodiversity conservation and sustainable use. While oceanographers in general, and marine biologists in particular, are aware of the challenges and complexities of measuring biodiversity in the ocean, the politics of marine biodiversity monitoring and the added value of social science perspectives tend to be neglected. This story from the frontline describes our experiences with turning marine biodiversity monitoring into an object of social science inquiry. The MARIPOLDATA project developed an innovative multi-sited approach to capture both the politics and practices of monitoring marine life. During our research, we were confronted with a diversity of perspectives and expectations of what monitoring is, how it works, why it should be done, and whom it should serve. By adopting a mixed-method approach and collecting data at different epistemic sites (negotiation rooms, scientific fields, laboratories, and conferences), we tried to cut through some of the observed ambiguities and trends. We conclude that studying monitoring necessitates a multidisciplinary approach and a sensitivity for the regional, institutional, and cultural specificities and inequalities that shape how we know and govern the ocean.

Keywords: marine biodiversity; monitoring; data; social science; ethnography; mixed-methods

Introduction

Marine biodiversity monitoring is high on the political agenda. From the Arctic Council to the OSPAR Commission and the Convention on Biological Diversity (CBD), the systematic observation and measurement of marine biodiversity change—its taxa, genes, structures, functions, and ecosystems—is fundamental to supporting decision-making. However, given that the ocean is a fluid, three-dimensional environment, it is not easy to know which species are present, where, and when, and to deduce related trends in biodiversity (Costello et al. 2017, p. 129). Nevertheless, while scientists acknowledge the biases and constraints of marine biodiversity monitoring (Danovaro et al. 2016, Levin et al. 2019, Danovaro et al. 2020, Danovaro et al. 2021, Ingels et al. 2021, Stefanni et al. 2022, McGeady et al. 2023), they do not to the same extent acknowledge the critical importance of the politics shaping the systematic observation and measurement of marine life or the added value of systematically studying related practices from a social science perspective.

Yet for social scientists looking into interrelations between science and policy, marine biodiversity monitoring in itself constitutes an object of research. Although this may be obvious for researchers working in the tradition of the social studies of science (Gabrys 2016, Ratte 2019), the history of science (Oreskes 2003), or geography (Lehman 2016, Peters and Steinberg 2019), what we study, how, and with what interest in mind may be less clear for researchers outside these fields. Hence, the UN Decade of Ocean Science explicitly values inter- and transdisciplinary perspectives (Visbeck 2018). It encourages ocean scientists to take the 'social factor' into consideration, both in their attempts to 'make oceanography fit for purpose' (Ryabinin et al. 2019) and to reach the objective of policy-relevant marine scientific research (Vadrot et al. 2022). Indeed, showing that political, social, and historical studies of science can serve science and policy to achieve better monitoring is becoming easier (Geijzendorffer et al. 2016). However, communicating why fundamental social science research is a precondition to fulfilling this reflexive role, and explaining what it is that we do when we study marine biodiversity monitoring is a challenging endeavour; this is particularly true in an ethical sense whenever ocean scientists (and their research practices) themselves become our research objects.

This story from the frontline demonstrates the added value of a social science understanding of marine biodiversity monitoring, arguing that it is a central component not only of marine scientific research but, also, of global environmental agreement-making and the struggle over environmental knowledge between the Global South and Global North (Vadrot 2020, Vadrot et al. 2022). We will introduce the MARIPOLDATA (www.maripoldata.eu) research project (2018–2024) and its unique multi-sited approach: in line with the idea of different geographies of knowledge (Mahony and Hulme 2018), it combines research at different epistemic sites: negotiation rooms, scientific fields, laboratories, and conferences. By combining research on the science and the politics of marine biodiversity, we seek to unpack what monitoring means in practice, why it has a political dimension, and what is at stake.

The interest of social scientists in marine scientific research is nothing new; it is rooted in several disciplines (Markus et al. 2017) and research traditions, including those that entail

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working closely with oceanographers to understand their day-to-day activities (Goodwin 1995, de Bont 2009). Historians of science, along with sociologists and Science and Technology Studies (STS) scholars, have sensitized us to the fact that ocean science is 'fundamentally social' (Mills 1993, p. 14; Steinberg 2001), and that 'situated knowledge' is produced in laboratories, in marine stations, and on research vessels through collective efforts and technological advances that, over time, have changed human ability to study marine life (Vermeulen 2013, Adler 2014, Lehman 2018, Bakker 2022). While in the early nineteenth century, individual natural philosophers could only study isolated elements of the sea, late twentieth-century technology and system-based approaches have made it possible to model change and build marine biodiversity scenarios in support of conservation measures. Using marine biodiversity monitoring data in decision-making is, however, hampered by many factors, including the difficulty of building (and agreeing on) indicators, integrating different data types, and addressing global inequalities within marine biodiversity research (Partelow et al. 2020). Consequently, infrastructures for 'data production, management, curation, and circulation' are key to overcome data gaps (Benson 2012, Darch and Borgman 2014, Drakopulos et al. 2022).

Marine biodiversity monitoring is not only about the four above-mentioned types of activity; it requires a multi-sited research approach that connects science and politics. The global politics of environmental data and knowledge are equally important; they include international efforts not only to further develop and harmonize methodologies, datasets, and indicators but, also, to attempt aligning these efforts with the needs of multilateral environmental agreements such as the CBD (Pereira et al. 2013, Proença et al. 2017, Schmeller et al. 2017). This is illustrated by the Kunming-Montreal Global Biodiversity Framework (GBF) adopted in December 2022 to make global biodiversity information relevant for national decision-makers (Buschke et al. 2023).

In writing this paper, we aim to broaden the understanding of marine biodiversity monitoring so as to include actors, sites, and processes at the interface between science and policy. We will take the reader to the sites that are relevant for the study of marine biodiversity monitoring from a social science perspective, before discussing some of the project's most relevant findings, and how they might contribute to interdisciplinary and transdisciplinary discussions on the future of monitoring marine life.

In the following, we will first introduce the two co-authors, their research interests, and their roles in the MARIPOLDATA project. We will then address the three challenges of studying marine biodiversity monitoring and demonstrate what social science perspectives can offer. We will adopt a multi-sited approach to cut through complexity and take the reader to the MARIPOLDATA research sites, reflect on our observations and findings, and show that our understanding of marine biodiversity monitoring developed throughout the research project. Finally, we will reflect on why the stakes of studying and enhancing marine biodiversity monitoring are so high. To conclude, in order to underpin future multidisciplinary research, we will offer a social science definition of marine biodiversity monitoring as a multifaceted set of physical, digital, and hybrid practices performed at the interface between marine scientific research, decision-making, and industry.

About the authors

As a principal investigator, Alice set up the MARIPOLDATA project in 2018, for which she obtained funding from the European Research Council (ERC). A political scientist and International Relations scholar with an interest in science-policy interfaces, Alice started her career with extensive ethnographic research on the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES). She developed the 'epistemic selectivity' concept, which she used to analyse the political struggle between state actors over the type of biodiversity data and knowledge that should support multilateral environmental negotiations (Vadrot 2014, 2016, 2020). In order to understand how this struggle relates to biodiversity knowledge production and use, she expanded the sites of her research beyond the negotiation room, thus including scientific practices and communities in the empirical study of science-policy interrelations.

Her first encounter with marine biodiversity monitoring communities was at the Group on Earth Observations (GEO)-BON conference in Leipzig, Germany, in June 2016. As a solo researcher, she conducted participant observation and interviews to study the interaction between two types of community: data scientists and 'field workers' (e.g. ecologists, taxonomists, and biologists). She decided to join the working group discussing the establishment of the MBON (Marine Biodiversity Observation Network) and observed the early formation of 'a "coalition of the willing" who agree to share knowledge and know-how to evaluate changes of biodiversity in the ocean, including data, products, protocols and methods, data systems, and software' (https://geobon.org/bons/thematic-bon/mbon/). In October 2017, she followed the group to the 14th Plenary of the Intergovernmental GEO, where they led a worldwide effort to build a Global Earth Observation System of Systems. As a result of her field observations, a large number of exploratory interviews with marine biodiversity researchers (e.g. Frank-Muller Karger, Patricia Miloslavich, or Isabel Sousa Pinto), and two workshops during her two-year postdoctoral research stay as an Erwin Schrödinger Fellow at the Centre for Science and Policy of the University of Cambridge (2015–2017), she designed the MARIPOLDATA project, for which she obtained ERC funding in 2018.

Krystel joined the MARIPOLDATA project as a postdoctoral researcher in early 2022 to coordinate the laboratory ethnography fieldwork and study monitoring communities at the sites where they meet to discuss their work. She spent three weeks at the IFREMER deep-sea laboratory (BEEP) in France and provided conceptual guidance to the two doctoral students in charge of ethnographic fieldwork in Brazil (Arne Langlet) and the USA (Ina Tessnow-von Wysocki). Krystel holds a PhD in Political and Social Sciences from the Université Laval and Free University of Brussels. Her work lies at the intersection between environmental politics, expertise, and international organizations. She has conducted extensive research on the United Nations Environmental Programme (UNEP) to apprehend its authority from the perspective of its experts' networks and careers.

Prior to joining MARIPOLDATA, she came across environmental remote sensing and monitoring communities during a side event of the 3rd United Nations Environmental Assembly in 2017 held on MapX, an open-source platform that makes it possible to map and monitor natural resources and the en-

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vironment using the best available and interoperable data. While studying the expertise related to the UNEP Global Resource Information Database, she encountered the challenges of monitoring the practices and policies that track the development of the UN Biodiversity Lab, which was created out of MapX to support the CBD's monitoring of national biodiversity strategies.

The challenge of (studying) marine biodiversity monitoring

'The lack of empirical biodiversity observation data is obvious at various levels of complexity; even basic inventories of current local-to-global biodiversity are missing. There are several reasons for this. Firstly, global cooperation in biodiversity research and monitoring is a relatively recent phenomenon. We lack standards, we do not yet share protocols, we do not consider strategic sampling, and there is limited exchange of data at and between spatial scales' (Jongman et al. 2017, p. 21).

This quote from Chapter Two of *The GEO Handbook on Biodiversity Observation Networks* exemplifies the challenges faced by scientists in monitoring marine life. Significant gaps in data or information and the search for meaningful, robust indicators make it difficult to measure change and build future marine biodiversity scenarios. Equally important, however, is the perception that 'global cooperation in biodiversity research and monitoring is a relatively recent phenomenon' (Jongman et al. 2017, p. 21) that requires social scientists dealing with marine biodiversity monitoring to make deliberate choices regarding their research focus, object, and methodology.

As regards turning marine biodiversity monitoring into an object of research, from a social science perspective, three developments are particularly interesting and challenging. The first one is rapid scientific and technological progress in the field: ocean monitoring capabilities have grown exponentially in the past decade to cover coastal areas, the high seas, seabed cable networks, and deep-seafloor imaging technologies. Even though they were initially intended for physical oceanography (Heesemann et al. 2014, Kelley et al. 2014), these systems (such as Argo floats or seafloor observatories) increasingly feed data into marine biodiversity monitoring, promising to contribute to a comprehensive description and understanding of processes taking place in the oceans (Levin et al. 2019). New detection methods, sampling regimes, 'big data' infrastructures, the digitization of historical data, and attempts to combine taxonomy, remote-sensing, and environmental DNA have created new approaches to the measurement of marine biodiversity change, the tracking of biodiversity monitoring, and modelling into the future (Costello and Chaudhary 2017, Turak et al. 2017). Digital infrastructures—used by vessels, marine station-based monitoring, robots, and remote sensors—have increased both the volume of data and the autonomy of monitoring systems (Lehman 2018).

In fact, global remote sensing observation systems have the potential to create a dynamic 'data double', a digital twin that holds the promise of real-time, mobile monitoring for conservation policies. Although vessels and human maintenance are still required to operate these monitoring systems, this more autonomous 'data setting' has increased the velocity and volume of the data flow and the dataset treated (Loukissas 2019). Social scientists argue that this constitutes a paradigm shift towards 'smart oceans' and digitally driven earth system governance induced by real-time monitoring (Bakker 2022). From this perspective, monitoring is not only about the data in the systems but, also, about the data products enabled by these systems, such as maps, stocks, forecasts, and records of endangered species and protected areas, and about 'dynamic ocean management' (Maxwell et al. 2015, Payne et al. 2017). Yet the effects of this paradigm shift are difficult to capture, even (and particularly) from an insider's point of view (Toonen and Bush 2020).

The second development results from the historically inherited fragmentation of monitoring infrastructures and the difficult task of bringing them all together (Muller-Karger et al. 2018). Countless efforts have been undertaken to coordinate data banks, platforms, and programmes such as the Global Ocean Observing System (GOOS) (Tanhua et al. 2019), the Marine Biodiversity Observation Network (MBON), and the Ocean Biogeographic Information System (OBIS). The latter began as a data repository for the Census of Marine Life (Grassle 2000) before expanding into a comprehensive platform for global marine biological observations fed by national repositories and a number of disparate data repositories. Both the spatial and temporal coverages of monitoring have also benefited from the production of globally and nationally relevant indicators, assessments, protocols, and principles that increase the sharing of data and methods (Stefanni et al. 2022). For instance, the Findable, Accessible, Interoperable, and Reusable (FAIR) principles (Wilkinson et al. 2016, Benson et al. 2018) delivered a roadmap for data management that facilitates data reuse and sharing. Efforts in steering monitoring practices through these infrastructures have provided guidance to scientists and managers while also helping to inform policymakers about the use of science and data (Patrício et al. 2016).

These data products also have implications for the emergence of new actors. Drakopulos et al. (2022) stress the role of data in monitoring, using the case of ocean data science initiatives (ODSIs) as a set of complementary schemes that support the advent of monitoring. An ODSI is an 'initiative that mobilizes (often geospatial and temporal) data and/or novel data sources about the oceans with the express goal of informing or improving conditions in the oceans'. Data are not only produced by actors who then use them; they possess an agency over actors producing and using them, creating, for instance, an oceanographic data 'market' far larger than that represented by professional oceanographers (Conway 2006). They are contentious and conflictual because some data products actually transform inert data into powerful instruments to govern oceans, such as maps. While social science scholars have studied how data is governed 'as norms and values about how they are accessed, stewarded, and applied are developed and shared', critical scholars amongst them have also highlighted the need to study how data are governing us through the rules that structure data governance, which are emerging from social, historical, political, and economic conditions (Drakopulos et al. 2022, Goldstein and Nost 2022).

The third development is the realization that marine biodiversity monitoring is inherently connected with policy (Wright et al. 2018). Ocean scientists often emphasize that they have a societal responsibility to present data objectively and call for dynamic interaction between science and policy (Lubchenco 1995, Levin and Le Bris 2015, Levin et al. 2019, Gaill et al. 2022, p. 8; Lubchenco et al. 2019, p. 107; Sullivan et al. 2017). The research community is meant to study the effectiveness of

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conservation measures by collecting baseline data and coordinating observation and monitoring activities. This involves the joint construction of indicators and variables (Malone et al. 2014, Muller-Karger et al. 2018) with an ethical stake (Barbier et al. 2018). The Essential Ocean Variables (Miloslavich et al. 2018) and marine Essential Biodiversity Variables (Pereira et al. 2013), which were conceptualized by the GOOS (Tanhua et al. 2019) and MBON, respectively, provide guidance for ocean observation; they were specifically designed to meet the needs of states in implementing the CBD and other related international environmental agreements. In this sense, marine biodiversity monitoring entails a dimension of stewardship, whereby the dynamic ocean space is constituted as a commons to be protected for future generations (Beckwith and Anderson 2017). Acknowledging the ocean as a commons creates a sense of responsibility and ownership amongst ocean scientists as regards why and how they observe and measure global biodiversity change. However, critical data studies have also emphasized the close relationship between monitoring and data-driven policymaking, whereby monitoring becomes a management tool, allowing humans to become planetary managers (Bowker 2000, Drakopulos et al. 2022).

As social scientists, we acknowledge that monitoring practices are transforming the relationship between humans and the ocean. Monitoring does something *to* the ocean and of the ocean (Bowker 2000, Devictor and Bensaude-Vincent 2016). It deepens the co-constitution of humans and the ocean—a more-than-human environment (Gabrys 2019) and active relational agent (Peters et al. 2022). In other words, human activities related to marine ecosystems and resources—their exploitation, conservation, sustainable use, and scientific study—are both shaped by and depend on the specific geological, chemical, physical, and biological composition of the ocean. At the same time, these activities are changing the ocean and what we know about it at a rapid pace, which makes it difficult to study natural and social processes in isolation from each other (Wanneau et al. 2023).

Consequently, marine biodiversity monitoring is materialized in concrete ways: observing marine life is inseparable from the ocean as a physical entity, but is also inherently connected to the technical, epistemic, political, and social relations that make the ocean legible. It is by taking these relational aspects into account that social scientists can contribute to uncovering barriers, inequalities, and selectivities; this required us—as part of the MARIPOLDATA project—to broaden the understanding of marine biodiversity monitoring by including actors, sites, and processes at the interface between science and policy. In the following, we will take the reader to the different epistemic sites of data collection and what we learned about marine biodiversity monitoring in theory and practice.

Cutting through complexity: a multi-sited research approach

It turned out to be more difficult than we expected to study marine biodiversity monitoring empirically. Ocean scientists themselves struggle to explain what monitoring entails when it is successful and why it sometimes risks failing. Data travel between political spaces and research sites, is enhanced, transformed, contested, and (mis-)used by diverse actors, including governments, international organizations, universities, industries, NGOs, and local communities (Conway 2006, Webster 2018, Loukissas 2019, Nost and Goldstein 2022). We tried to cut through these ambiguities and study the politics and practices of monitoring marine life by acknowledging the importance of different epistemic sites and geographies of knowledge (Mahony and Hulme 2018), all the way from the negotiation room to the laboratory. In a nutshell, our approach combines ethnographic research during global environmental negotiations and agreement-making (Hughes and Vadrot 2023a, Hughes and Vadrot 2023b) with a systematic social science analysis of scientific fields, stories, and practices (see Vadrot 2020, Vadrot et al. 2022).

We start from the assumption that global environmental negotiations are inherently connected to how we know, measure, monitor, and govern biodiversity in the ocean and on land. During environmental negotiations, states engage in law-making and increasingly explicitly determine the role and importance of monitoring in implementing and reviewing conservation measures. However, instead of focusing on the negotiation room only, we broaden related sites to 'agreement-making'. This implies that we consider 'the multiple actors, sites, and processes through which environmental agreements are made, and the new sets and arrangements of actors, sites, and processes that are created by any specific agreement, which have the potential to reinforce or reorient the global political order' (Hughes and Vadrot 2023b, p. 1). Those include scientific fields and practices demanding an additional set of methods to complement data collected during negotiations (on the role of science, data, and monitoring) with analyses of the conditions for producing marine biodiversity data and knowledge. We mapped the scientific field of marine biodiversity research by using bibliometric analysis (Tolochko and Vadrot 2021a, 2021b), collected stories of marine biodiversity scientists by conducting oral history interviews, and observed day-to-day interactions between different types of actor during scientific conferences and in marine laboratories.

The next section provides an account of what we observed at these epistemic sites and illustrates the iterative process that resulted in our definition of monitoring as *a multifaceted set of physical, digital, and hybrid practices performed at the interface between marine scientific research, decision-making, and industry.*

In the negotiation room

We started our research journey in the spring of 2019 in Conference Room No. 4 at the United Nations' headquarters in New York. We were observing proceedings related to the so-called 'Biodiversity Beyond National Jurisdiction (BBNJ) treaty'—a legally binding instrument for the sustainable use and conservation of biodiversity in areas beyond national jurisdiction (ABNJ) that was meant to fill legal gaps in the Law of the Sea (see Fig. 1). Our collective observation of how over one hundred governments negotiated the future of marine life in the multilateral setting of environmental diplomacy offered windows into the meanings attributed to marine biodiversity monitoring and research both by state and non-state actors (Tessnow-von Wysocki and Vadrot 2020, Vadrot 2020, Langlet et al. 2022). We analysed the struggle over scientific concepts such as 'ecological connectivity' (Tessnow-von Wysocki and Vadrot 2022) and legal concepts such as the 'common heritage of humankind' (Vadrot et al. 2021) in order to better understand the interests and values of actors in relation to marine biodiversity conservation and sustainable use. By en-

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Figure 1. BBNJ Negotiations, Informal Working Group on Cross-Cutting Issues, 5th April, 2019, Earth Negotiation Bulletin (ENB) Report, retrieved from https://enb.iisd.org/events/2nd-sessionintergovernmental-conference-igc-conservation-and-sustainable-use-marine-9 on 12.03.2024.

tering this UN site, we were able to study the effects of legal developments on science—a much-needed research effort to clarify science-policy interactions (Vanderpool 1983).

The underlying politics of marine biodiversity monitoring in this setting became apparent when governments discussed whether to regulate access to and use of marine genetic resources (MGRs) in ABNJ and, if so, how. MGRs in the high seas are considered a common good but, increasingly, are being exploited by a handful of biotech companies with an interest in those genetic properties of biodiversity that generate economic benefits through patenting marine life (Blasiak et al. 2016, Blasiak et al. 2018). Many states, especially from the Global South, viewed monitoring as a precondition to benefit-sharing and reverting the trend towards ocean privatization, rather than as a tool to monitor the impact of Marine Protected Areas (MPAs) or assist with Environmental Impact Assessments (EIAs). We observed similar dynamics during CBD negotiations on the GBF during COP 15 (December 2022). Global South states pointed to the need for more financial support to monitor marine biodiversity in their waters; they viewed this as a precondition to measuring implementation progress and complying with the new agreement.

Diving into intergovernmental negotiation processes, we found that the politics of marine biodiversity monitoring centred around the tensions between the industrialized world and developing countries as regards their monitoring capacities and access to marine resources. Inside the negotiation room, the capacity to monitor marine biodiversity was turned into an essential tool to gain privileged access to marine life and exercise power over the marine world. One example is the discussion over the Common Heritage of Humankind, where unequal access to the ocean as a space to be explored and monitored played an essential role (Vadrot et al. 2022). Another example relates to the potential of tracking MGRs (from the collection of a water sample to the final product), which some state actors equally associated with what monitoring systems in the BBNJ context should do. More importantly, monitoring was considered to be a decisive factor that might determine the future of global environmental diplomacy and the ability of the international community to govern marine biodiversity collectively and in line with the best available science. For many scientific actors in the room, this was especially relevant when it came to the designation and management of MPAs and making EIAs. In order to understand how tensions over the meaning of monitoring and the unequal distribution of monitoring capacities relate to dynamics within the scientific field and scientific collaboration patterns between countries, in parallel to this exercise in Collaborative Event Ethnography, we started mapping the scientific field.

Mapping the scientific field

As a multifaceted set of practices, marine biodiversity monitoring is a central element of marine biodiversity research and the scientific study of marine life. Drawing on the idea of 'world-building' (Callon et al. 1986), we analysed the dynamics of the marine biodiversity field, our specific aim being to identify regional and temporal specificities regarding research topics (Tolochko and Vadrot 2021a, 2021b) and to challenge the paradigm and imaginary of a globalized marine biodiversity 'big science' (Price and de Solla Price 1986): 'what exactly constitutes the field of marine biodiversity in terms of content,

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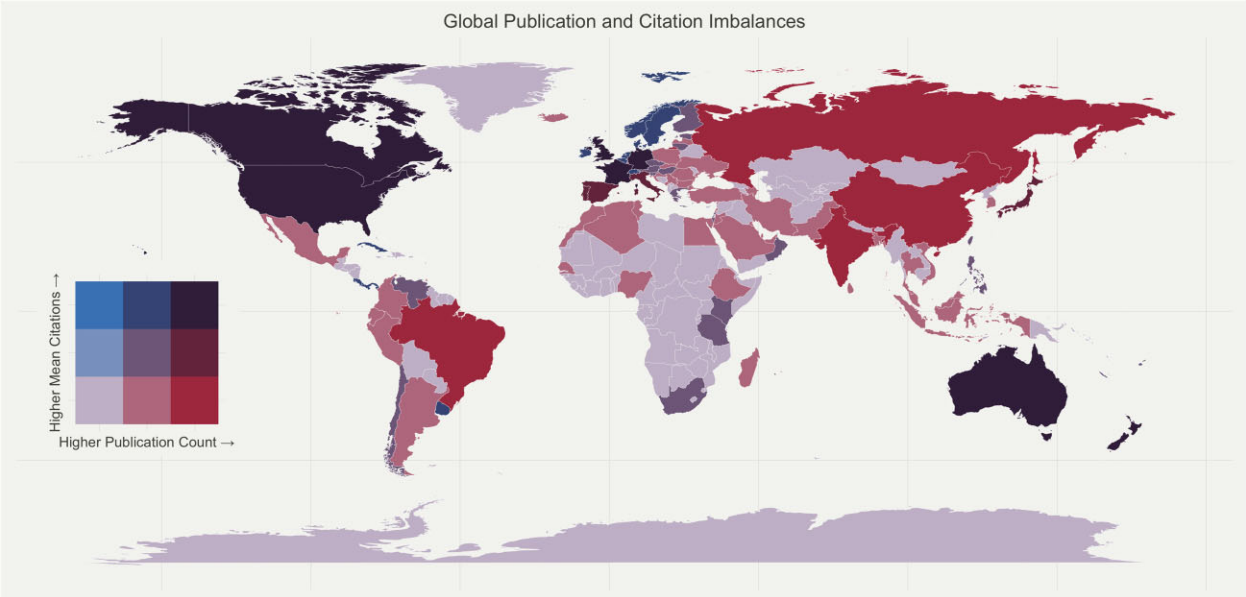

Figure 2. Geographic distribution of the total amount of articles and average citation count by country, 1990–2018 (source: Tolochko and Vadrot 2021a published in Marine Policy).

[…] what regions play what role in the production of scientific knowledge within the field' (ref. missing) and what is the effect of international collaboration on the thematic focus of research in different regions of the world?

An emerging field at the intersection between ocean science and environmental science (Cochrane et al. 2016), marine biodiversity research is dependent on large-scale international research programmes and observing systems, as well as international collaboration and large-scale investment into research and data infrastructure by governments and private donors (Tolochko and Vadrot 2021b). Our bibliometric analysis of >24 000 publications confirmed what developing states pointed to during BBNJ negotiations: a few countries from the Global North—the 'usual suspects'—dominate the field (see Fig. 2).

By investigating its composition and the regional distribution of research topics, we gained a more nuanced understanding of what marine biodiversity research is about and why marine biodiversity monitoring matters (see Fig. 3). Research into genetic diversity seems to be preeminent, with three clusters related to (1) populations and species descriptions; (2) digital sequence information and product development; and (3) sequence information in taxonomy and for the classification of species. The field, however, is equally concerned with methodological considerations (how to assess and how to map) and data collection techniques, *inter alia*, to demonstrate that marine biodiversity loss is actually real and to develop a comprehensive understanding of what conservationists are hoping to protect.

In many Latin American countries, for instance, classical taxonomy is much higher on the research agenda—as we learned later by talking to scientists. Indeed, Latin American taxonomists are often considered world leaders; they are masters of a method assessing the risk of extinction, a practice now often overtaken by advances in computer science, 'big data' technology, and modern taxonomy modelling methods fostered by the DNA revolution in the 1990s and camera technology innovation in the 2000s (Bicknell et al. 2016). The latter development suggests that we are very close to obtaining a global coverage of species and a full picture of *what* marine biodiversity is composed of, *where*, and *why*. While our bibliometric data suggests that some methods and approaches are more influential in some regions than in others—except for genetic research, which has dominated the field of marine biodiversity in the past decade—we could only grasp the nuances and what this means in practice by interviewing marine biodiversity researchers.

Collecting oral histories

Generally speaking, Oral History of Science helps us 'document important dimensions of the history of twentieth-century science, mostly from the scientists' point of view' (Weiner 1988). By conducting history of science interviews during our project, we intended to collect subjective views, narratives, and memories of key researchers constituting the marine biodiversity field. This is a field in its own respect—somewhere between oceanography, biological oceanography, and marine biology—as we learned by listening to those that 'do science' (Weiner, 1988).

While preparing the interviews and developing the MARIPOLDATA approach to Oral History of Science, we learned that oceanography had been studied as one of the first 'big sciences' (Mills 1993, Reidy 2009). In fact, this field, as no other, has been dependent on a convergence of ships, equipment, and instruments (Mills 1993); no other field has 'been so dependent on the growth of the modern state' (Burstyn 1968a, p. 670) or 'has taken shape within particular conceptions of place, interests, and sovereignty—not least because the ocean sciences are involved with the practical and epistemological control of the earth's resources and are therefore intimately connected with the realities of the nation-state and its institutions, on the one hand, and a globalized economy, on the other hand (Rozwadowski 2014, p. 336).

Adopting a historical perspective when listening to key scientists from the marine biodiversity field turned out to be im-

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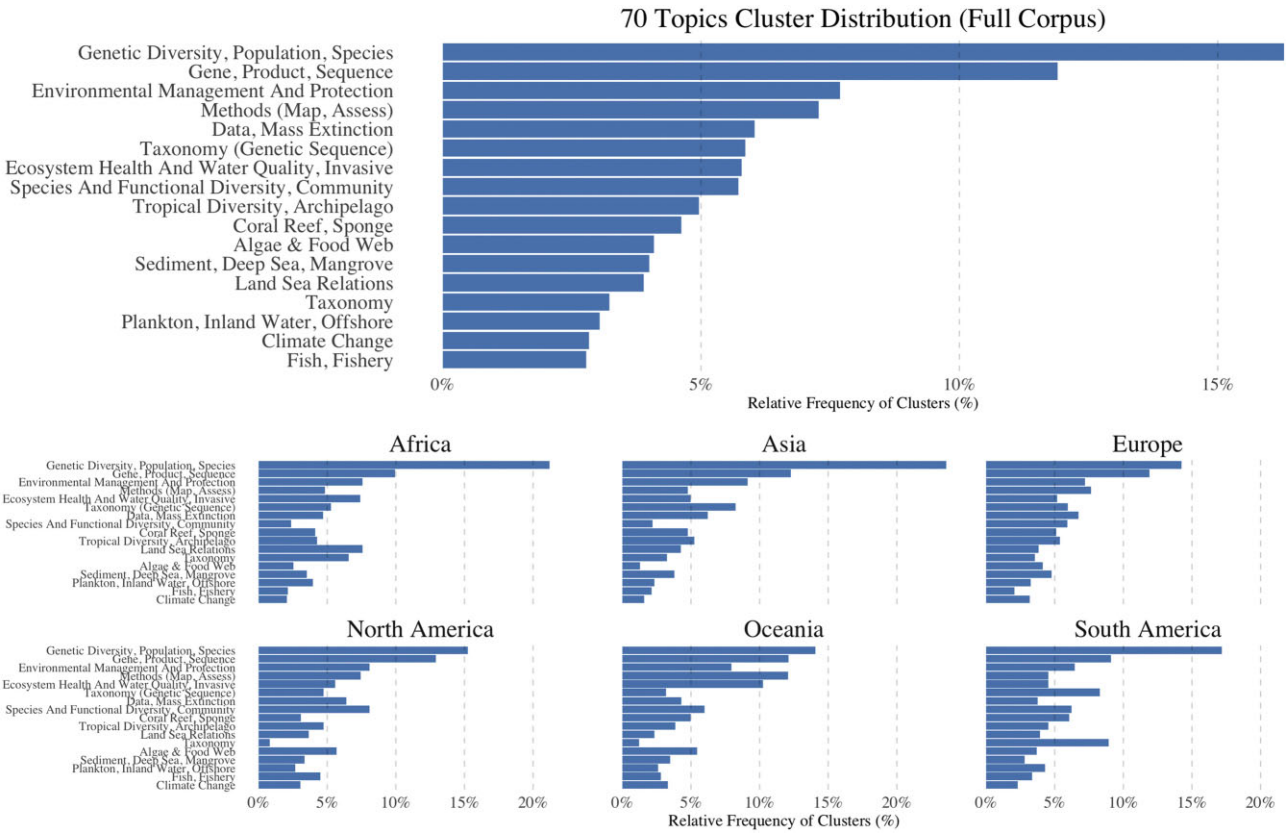

Figure 3. Relative Frequency of Clusters. Top panel shows the relative frequency of clusters in the whole dataset. Bottom panel shows the relative frequency of clusters broken down by regions (source: Tolochko and Vadrot 2021b).

portant in many ways. It showed us that marine biodiversity research and monitoring are actually 'much older' than our bibliometric data sample suggested. While the term 'biodiversity' was only coined in the 1980s, the quest to study, describe, and classify marine life is much older; indeed, many of our interviewees see themselves as being part of a longstanding tradition of scholars exploring nature and life forms because they care about the natural environment and its beauty (Worster 1994). Interviewees referred to ancient philosophers, including Aristoteles and Heraclitus, but also to Captain James Cook (1728–1779), Charles Darwin (1809–1882), Sir Charles Wyville Thomson (1830–1882), or Jacques Cousteau (1910–1997).

Historically, ocean science was closely connected to the modern state's desire to enclose the marine realm (Rozwadowski 2014, p. 336), a desire that has spread to many private actors trying to fill the gap. The mission pursued by naturalist exploration, collecting species, which was once funded by states, has shifted in part to philanthropic organizations. Besides valuable, state-led ocean DNA sampling initiatives such as the Tara expeditions, these organizations? are producing new forms of (digital) mediation that alter or complicate the historical narrative; one interviewee was critical:

> *There's more need than ever for more marine scientists to be out there collecting samples, monitoring what's happening and I don't think we're doing it. We're missing the opportunity there. That's why you do see these philanthropic organizations stepping up. Now you can see the FalKor research ship run from California by that philanthropic organization. It's a private research ship, and you can apply for ship time on it. Really? Should that really be? Shouldn't countries be doing this? Why are we relying on philanthropists? That's what's happening. There's been a lot of advances in some areas and yet there's still a lot we could be doing. We could do better. (OHI, 19.07.2021)*

Some of the scientific breakthroughs in the study and monitoring of marine life—thanks to remote sensing, environmental DNA, and artificial intelligence—hold promises that have fed technological imaginaries, including the aforementioned transition to 'smart oceans'. These, however, do not match the research realities of many marine biodiversity scientists struggling in their day-to-day work. By interviewing them, we identified a series of gaps between data infrastructures (as envisioned) and monitoring practices that might impede/hamper/slow down the systematic, real-time observation of marine life around the globe? While these gaps do not prevent the emergence of monitoring systems, they do create a general sense of disorder that undermines the intelligibility of these human-technology-environment interactions, as illustrated in the quote below related to data management:

> *[…] actually, it's very badly managed. Everything is in tapes, like CDs, DVDs, it's a mess. So like a year ago, we were getting information to make this more available in some format or in a databank. But now, it's like a mess. We requested them, they said you have to send a person here to see what we have, and to transform that, to put that in some database or something, but it should be better. But unfortunately, it's not. (Oral History Interview, 18.08.2022)*

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In order to observe practices, we needed to go beyond the subjective stories of marine biodiversity researchers and visit the sites where monitoring communities meet (events) and where monitoring actually takes place (the laboratory).

Encounters with monitoring communities 'on stage'

Our bibliometric research and oral histories suggest that monitoring practices are messy and undergoing rapid transformation owing to digitization, and that they are prone to failure. However, what should we as social scientists 'observe' when we try to 'see' and study monitoring in the making? Following the path opened up by marine biodiversity researchers' stories and memories, we started to engage with monitoring communities during three types of events: the EOOS Technology Forum, the One Ocean Summit, and the 4th Marine Imaging Workshop (see Table 1).

At the One Ocean Summit, organized in the context of the French Presidency of the Council of the European Union, and with the support of the United Nations, we could observe monitoring communities at the highest political level. The summit offered an on-site and online platform for policymakers, scientists, representatives of civil society, and business people to discuss future ocean science and governance issues. In addition to side events and smaller workshops, policymakers (most notably the President of the European Commission, Ursula van der Leyen) advocated for visionary ways to study and represent the ocean by developing digital ocean twins (i.e. real-time representations of the ocean). We could sense the paradigm shift discussed in the literature and the urgency of finding solutions to close data gaps and provide policymakers with readily available knowledge. Speeding up data collection—with the help of new devices, modelling, and artificial intelligence—was presented as the way forward to prevent further damage to the marine environment, a precondition being close cooperation between providers and users of data and technology.

We observed a similar convergence of public and private actors at the EOOS Technology Forum, where participants delved into monitoring system innovations related to data infrastructures, underwater vehicles, sensor devices, and satellites. The Forum offered technology users, developers, and providers a platform to jointly develop technology. During the Marine Imaging Workshop in October 2022 (see Fig. 4), we noticed that these developments could also create friction and lead to conflict. For instance, researchers disagreed on the data types that would be needed to monitor marine biodiversity, as well as on what the paradigm shift described earlier would mean in practice—in particular, for more 'traditional' data collection techniques such as classical taxonomy.

Thanks to our participation in the three events, we could see the paradigm shift happening across scientific, political, and private spheres; moreover, we could sense the emergence of new actors and interest groups. We identified a number of knowledge entrepreneurs who attended these events to promote their tech solutions to monitoring challenges and their new tools, such as FathomNet, an open-source image database designed to enable AI in oceanographic monitoring (Katija et al. 2022), and FAIR principles for underwater images intended to facilitate image publication in long-term repositories (Schoening et al. 2022). Our observations gave us insights into uncertainties, opacities, and, sometimes, resistance both from within and outside the laboratory (e.g. regarding multilateral agreements, national regulations, funding schemes, etc.). However, in order to understand what this meant in practice, we needed to turn our attention to the sites where monitoring plays a central role in the day-to-day work of researchers: laboratories.

Observing monitoring practices in the laboratory

To observe monitoring practices in the making, we immersed ourselves in laboratories. In so doing, we followed a long tradition of laboratory studies (Latour and Woolgar 2006), including in marine stations, where ethnographic studies have already been conducted to understand daily lab practices and how these produce knowledge about the deep sea (Goodwin 1995) or wildlife biology (Benson 2012). Fieldwork preparation and gaining access to the relevant laboratories took us six months. Then, we conducted three two-week-long ethnographic studies in France, Brazil, and the United States; we also participated in the day-to-day work of three deep-sea laboratories specialized in marine biodiversity and biology: in Brest, at the French national institute for sea exploration and exploitation (IFREMER); in São Paulo, at the University of São Paulo; and in San Diego, at the Scripps Institute. We attended research activities and were all very warmly welcomed, both before our arrival and during the actual work at the site.

To coordinate the collective laboratory ethnographic fieldwork, we developed a common strategy for the study of social interactions, research infrastructures, and the epistemic production of knowledge (Stephens and Lewis, 2017). We also created data collection guidelines to harmonize observations and facilitate comparison. During the fieldwork, we had daily online meetings with one another, recorded the visual data, conducted interviews with lab members (technicians, scientists, or engineers), and reported observations in a digital ethnographic diary.

Our findings suggest that laboratories are involved in marine biodiversity monitoring through three types of roles and activities: (1) as infrastructure caretakers, laboratories maintain monitoring infrastructures such as deep-sea observatories and landers; (2) as data curators, they provide the necessary environment for data collection, curation, and integration into repositories; and (3) as epistemic innovators, they are central sites where new monitoring protocols are developed and tested. However, the making of protocols and their inclusion in monitoring systems are far from being a linear process; this is something we noticed while observing the development and use of protocols for benthic meiofauna identification by the Blue Revolution (BlueREV) project at the IFREMER deep-sea laboratory (BEEP).

Through interviews with key researchers, we learned that in 2020, the BlueREV research team had borrowed a confocal-microscopy-based protocol developed for plankton during the Tara Ocean expedition and modified it for the purpose of rapid taxonomic recognition of benthic meiofauna organisms. Thus, they adapted a protocol dealing with coastal biodiversity to deep-sea observation conditions by applying technological innovations from other fields. To this end, they combined the expertise of the various project partners: taxonomic knowledge about nematodes and microscopy techniques located inside the BEEP lab (Kitahashi et al. 2018; Zeppilli et al. 2018; Zeppilli et al. 2015).

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Table 1. Main features of each event we participated in.

Title of event	One Ocean Summit	EOOS Technology Forum	4th Marine Imaging Workshop (MIW)
Type of event	Policy summit	Technology workshop	Scientific workshop
Date	9–11 February 2022	22–24 March 2022	3–6 October 2022
Organizer	French presidency of the Council of the European Union and the United Nations	EuroGOOS	Deep-sea laboratory of IFREMER
Purpose	'A Summit to take action against the threats to our Ocean'	'Thinking Ahead: The Technology of the Science We Will Need for the Ocean We Want'	'Push the boundaries of marine imaging'
Community	'European and international community'	'Ocean observing community', European Ocean Observing System (EOOS)	'Global imaging community'
Participants	41 states, and representatives of civil society and business	Technology users, developers, and providers	Multidisciplinary scientists and engineers

Figure 4. Group picture of the 4th Marine Imaging Workshop (source: https://miw2022.sciencesconf.org/).

Observing the everyday work of the laboratory, we could understand what was happening inside and outside the research facilities to create the new monitoring protocol. We joined the team to collect testing samples nearby the Roscoff marine station. They separated fauna from sediment in their storage lab (a slow process) and then applied the FlowCAM confocal imaging technique in another IFREMER lab. In order to transform the existing protocol to meet the needs of deep-sea observation, the team also borrowed expertise outside the lab from fields such as computer, physics, and optical sciences and tested new techniques outside their own lab facilities. Firstly, they collaborated with AI experts from the Laboratory of Sciences and Techniques of Information, Communication, and Knowledge (Lab-STICC). With their support, they combined public and private image datasets to train machine-learning models and algorithms to recognize nematodes. Secondly, to test the *in situ* digital holographic camera (HoloCAM) protocol, the BlueREV team collaborated with the Detection, Sensors, and Measurements Laboratory (LDCM) based at the R&D department of IFREMER. Their engineers upgraded the HoloCAM system acquired from Sequoia, a Californian start-up that invented the Laser In Situ Scattering and Transmissometer (LISST) aquatic science sensor series. Before the BlueREV team could send the HoloCAM to the deep sea, coupled with an artificial intelligence (AI) fast-recognition and automated classification programme on-board of the vessel, the LDCM team conducted several testing phases, including calibration conditions and the validation of the new monitoring protocol. Our observations conclude that one central activity, testing, not only involved internal laboratory life but, also, partners from other labs and the industry, as in the case of the above-mentioned start-up.

Studying the making of a protocol through participant observation and interviews offered new perspectives on monitoring and the connections between several sets of practices, technologies, and scientific disciplines. We found that a broad range of experts and institutions must perform various tasks both inside and outside the laboratory to develop baseline data used to observe and measure marine life. While far-reaching technological innovation affects monitoring, it has to coexist with traditional sampling practices and taxonomic expertise. Adapting protocols to technological innovation such as AI implies changing observation and measurement practices, but only to the extent that it does not impede the comparability of data and long-term monitoring objectives. This tension was evident when we reflected on what global monitoring infrastructures and data repositories, such as GOOS or OBIS, would require as regards data curation, standards, and interoperability, and what this would mean for the day-to-day work of those who monitor marine life.

Our observations in three marine laboratories revealed that the paradigm shift towards 'smart oceans' mentioned above

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is real but not straightforward; nor is it devoid of potential conflicts over practices in the race to find solutions fulfilling new monitoring system requirements. How 'smart oceans' enter the laboratory therefore depends on cultural, institutional, and regional specificities, the purpose of marine biodiversity monitoring, how it adapts to novel techniques (Matabos et al. 2022; Stefani et al. 2022), and the diverse needs of stakeholders.

High stakes: better monitoring for better decision-making?

During our research journey, we noticed that the closer we were to an understanding of marine biodiversity monitoring, namely of how, why, and where it is practiced, and by whom, the less we seemed to know what it actually is—a feeling that we even shared with those who engage in it on a day-to-day basis. Technological innovations, fragmented systems, competition over national and global funding, and competing policy needs and interests challenge marine biodiversity monitoring and the social study thereof. We experimented with social science methods, moving closer to, although not fully grasping, this messy *assemblage* of physical, digital, and hybrid practices that is inherent to how we observe and measure marine biodiversity change. However, the stakes for improving marine biodiversity monitoring to serve decision-making are high.

Beginning in the negotiation room, we witnessed the entanglement of marine biodiversity monitoring with policy. The BBNJ treaty negotiations and CBD COP 15 revealed the urgent need for marine biodiversity monitoring to support the objective of protecting 30% of the oceans by 2030. Many global environmental agreements are now at the stage of the implementation, which implies pressure on governments to comply and to report to the international community on whether and how they have reached global sustainability goals. Knowledge-supporting mechanisms, along with scientifically robust and politically legitimate indicators, will be essential to measure progress and provide governments with the information needed to decide where to establish a MPA or how to interpret an EIA, more so in the high seas where a global commons is at stake.

However, during our research journey from the negotiation room to the laboratory and the deployment of protocols, we encountered several challenges that those involved in monitoring are well aware of: global inequalities; regional, institutional, and cultural specificities; fragmentation; funding constraints and competition; interoperability issues; and persistent data gaps. Global visions of how monitoring should be conducted can conflict with local and regional monitoring practices, priorities, and objectives. By mapping the field of marine biodiversity research, we could detect temporal and regional specificities and, thereby, anticipate the long way we will need to go to fill knowledge gaps in many parts of the Global South. Our oral history interviews revealed the historically inherited fragmentation of monitoring infrastructures, the unequal effects of expanding 'smart oceans', and how they affect the scientific field, research careers, competition between research methods and approaches (e.g. classical taxonomy or eDNA), and monitoring practice. Marine scientists are experiencing rapid scientific and technological developments in the field, and feel under pressure to make their knowledge relevant and usable for decision-making. Indeed, the pressure to accelerate data availability and usability through digitization and artificial intelligence drove the three events that we attended to encounter monitoring communities, confirming what we identified in the literature as a paradigm shift towards 'smart oceans'.

In order to unpack how this globalized technology-driven vision of marine biodiversity monitoring affects current monitoring practices, we had to turn our attention to the sites where marine biodiversity is actually observed and measured. Laboratories gave our team access to a multifaceted set of practices, allowing humans to study marine life systematically. By developing methods and protocols to study unknown segments of marine biodiversity, such as benthic meiofauna, laboratories are sites where monitoring practices diversify and proliferate. In laboratories, diverse research teams with different disciplinary backgrounds innovate protocols and workflows, use and take care of monitoring infrastructures, and perform data-curating practices underpinning global monitoring systems.

Unlike what we expected when gaining access to the actual practice 'on the ground', marine biodiversity monitoring is not a list of practices that come together to produce 'the' activity of monitoring. The political struggle over environmental knowledge and what counts as a usable and reliable evidence base—to evaluate trade-offs between different forms of marine resource and ecosystem use—is, indeed, not detached from the challenges that researchers may encounter during their day-to-day work, especially if they feel that they have a stake and would like their research to support conservation efforts. High stakes can encourage new alliances and actor constellations, both in the negotiation room and the laboratory, where political and economic interests might converge; on the other hand, they can also create new forms of competition whereby public and private actors vie for data access and ownership. The emergence of digital ocean twins and the increased use of AI may well exacerbate this situation.

To avoid 'epistemic selectivity' (Vadrot 2014) as regards how, why, and under which institutional conditions marine biodiversity monitoring should serve decision-making, some sensitivity to the temporal, regional, and cultural specificities and inequalities across scales and sites is needed. For this reason, we adopted a multi-sited research approach, moving across sites where monitoring is negotiated, promoted, and performed. Broadening the understanding of marine biodiversity monitoring and arguing that it is a central component not only of marine scientific research but, also, of global environmental agreement-making, offers new research avenues; for instance, one might combine political science and international relations research with insights and methods from STS and critical data studies. To facilitate multidisciplinary research, we propose to define marine biodiversity monitoring as a multifaceted set of physical, digital, and hybrid practices performed at the interface between marine scientific research, decision-making, and industry.

Future research will have to go a step further by engaging with stakeholder needs and diving deeper into the actual use of marine biodiversity monitoring in policy, business, and industry. Not only might this increase the legitimacy, saliency, and credibility of existing monitoring systems but it would also support more inclusive, diverse, and equitable forms of knowing and governing the (smart) ocean.

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Author contributions

Alice B.M. Vadrot: Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Writing – original draft, Writing – review & editing

Krystel Wanneau: Investigation, Conceptualization and Methodology (WP4), Writing – original draft

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